

```
def greeting():
    a = "Good Morning"
    return a

print(greeting())
print(type(greeting()))
```

Good Morning
<class 'str'>

```
def summofothree():
    a,b,c = int(input()), int(input()), int(input())
    return a+b+c

print(summofothree())
```

3
4
5
12

```
def fact(n):
    if n == 0:
        return 1
    else:
        return n * fact(n-1)

n = int(input("Enter a num for fact: "))
a = fact(n)
print(f"{n}! = {a}")
```

```
def sumofnum():
    a = list(map(int, input().split()))
    return sum(a)
print(sumofnum())
```

```
def calculate_of_three(a, b, c):
    addition = a + b + c
    subtraction = a - b - c
    multiplication = a * b * c
    power = a ** b
    div = a / b

    return addition, subtraction, multiplication, power, div

a, b, c = map(int, input().split())

print(calculate_of_three(a, b, c))
```

```
def isprimenumber(n):
    for i in range(2, n):
        if n % i == 0:
            return 0
    return 1
a = int(input("Enter a num: "))
print(isprimenumber(a))
```

```
def primeinarange(m, n):
    a = []
    for i in range(m, n+1):
        if i > 1:
            for j in range(2, i):
                if i % j == 0:
                    break
            else:
                a.append(i)
    return a

print(sum(primeinarange(2, 10)))
print(primeinarange(2, 10))
```

```
def evenoddinarange(a,b):
    even = []
    odd = []
    for i in range(a, b+1):
        if i % 2 == 0:
            even.append(i)
        else:
            odd.append(i)
    return even, odd

print("Even List:")
print(evenoddinarange(1, 30)[0])
print("Odd List:")
print(evenoddinarange(1, 30)[1])
```

```
def expo(a,b):
    a = int(input("Enter a num: "))
    b = int(input("Enter a num: "))
    ans = 1
    for i in range(b):
        ans = ans * a
    return ans

print(expo(a,b))
```

```
a = list(map(int, input().split()))
k = 5

for i in range(len(a) - k + 1):
    w_sum = 0
    for j in range(i, i + k):
        w_sum += a[j]
    print(w_sum)
```

```
def haspair(arr,t):
    left=0
    right=len(arr)-1
    while left<right:
        s=arr[left]+arr[right]
        if s==t:
            return True
        elif s<t:
            left+=1
        else:
            right-=1
    return False
print(haspair([1,2,3,4,5,6],10))
```

True

```
def sqandsqroot(n):
    sq = n ** 0.5
    sqroot = n ** n
    return sq, sqroot

x = int(input("Enter a num: "))
print(sqandsqroot(x))
```

Enter a num: 16
(4.0, 18446744073709551616)

```
def addtwo(a,b):
    added = a + b
    return added

x, y = map(int, input().split())
print(addtwo(x,y))
```

3 4
7

```
def first(a):
    a = input()
    return a

print(first(a))
```

```
Hello'
Hello'
```

```
def areaoftriangle(l,b):
    area = 0.5 * l * b
    return area

#l = int(input("Enter a num: "))
#b = int(input("Enter a num: "))
print(areaoftriangle(l,b))
```

```
15.0
```

```
def linear_search(a, target):
    for i in range(len(a)):
        if a[i] == target:
            return f"Found at {i}"
    return "Not Found"

a = list(map(int, input().split()))
target = int(input("Enter a num: "))
print(linear_search(a, target))
```

```
4 3 5 6 4
Enter a num: 6
Found at 3
```

```
def quadeq(a,b,c):
    a = int(input("Enter a num: "))
    b = int(input("Enter a num: "))
    c = int(input("Enter a num: "))
    d = (b**2) - (4*a*c)
    if d > 0:
        print("Roots are real and unequal")
    elif d == 0:
        print("Roots are real and equal")
    else:
        print("Roots are imaginary")
    return d

print(quadeq(5,6,7))
```

```
Enter a num: 4
Enter a num: 5
Enter a num: 1
Roots are real and unequal
9
```

```
def swapwithc(a,b):
    a = int(input("Enter a num: "))
    b = int(input("Enter a num: "))
    c = a
    a = b
    b = c
    return a, b

print(swapwithc(a,b))
```

```
Enter a num: 5
Enter a num: 4
(4, 5)
```

```
def swapwithoutc(a,b):
    a = int(input("Enter a num: "))
    b = int(input("Enter a num: "))
    a, b = b, a
    return a, b

print(swapwithoutc(a,b))
```

```
Enter a num: 5
Enter a num: 4
(4, 5)
```

```
def swap(a,b):
    a = int(input("Enter a num: "))
    b = int(input("Enter a num: "))
    a = a + b
    b = a - b
    a = a - b
    return a, b
```

```
print(swap(a,b))
```

```
Enter a num: 5
Enter a num: 4
(4, 5)
```

```
import random as rd
def randomgen(a):
    a = rd.randint(0,10)
    return a
print(randomgen(a))m
```

```
9
```

```
#1 km = 0.6 miles
```

```
def distance(d):
    d = 0.6 * d
    return d
```

```
n = float(input())
print(distance(n))
```

```
23.23
13.938
```

```
#1mile = 1.6 kms
```

```
def distance(d):
    d = 1.6 * d
    return d
```

```
n = float(input())
print(distance(n))
```

```
13.938
22.300800000000002
```

```
def celtofah(t):
    t = (9/5) * t + 32
    return t
print(celtofah(50))
```

```
122.0
```

```
def fahtocel(t):
    t = (5/9) * (t - 32)
    return t
print(fahtocel(122))
```

```
50.0
```

```
def sign(n):
    if n > 0:
        print("Positive")
    elif n < 0:
        print("Negative")
    else:
        print("Zero")

m = int(input())
sign(m)
```

4
Positive

```
def evenodd(m):
    if type(m) == int:
        if m % 2 == 0:
            print("Even")
        else:
            print("Odd")
    else:
        print("Invalid Input")
n = int(input())
evenodd(n)
```

4
Even

```
def BinarySearch(a,high,low,target):
    while low <= high:
        mid = (high + low) // 2
        if a[mid] == target:
            return mid
        elif a[mid] < target:
            low = mid + 1
        else:
            high = mid - 1
            return - 1
a = list(map(int, input().split()))
target = int(input())
print(BinarySearch(a,len(a)-1,0,target))
```

3 4 5
4
1

```
def year(y):
    if y % 4 == 0:
        if y % 100 == 0:
            if y % 400 == 0:
                print("Leap Year")
            else:
                print("Not a Leap Year")
```

```
z = int(input())
year(z)
```

2026

```
def year(y):
    if y % 4 == 0:
        if y % 100 == 0:
            if y % 400 == 0:
                print("Leap Year")
            else:
                print("Not a Leap Year")
        else:
            print("Leap Year")
    else:
        print("Not a Leap Year")
```

```
z = int(input())
year(z)
```

2028
Leap Year

```
def largest(a,b,c):
    if a > b and a > c:
        print(a)
    elif b > a and b > c:
        print(b)
    else:
        print(c)
```

```
v v 7 = map(int, input().split())
```

```
3 4 5
5
```

```
def largest(a):
    big = a[0]
    for i in range(1, len(a)):
        if a[i] > big:
            big = a[i]
    return big

a = list(map(int, input().split()))
print(largest(a))
```

```
4 3 5 6 7 8 9
9
```

```
def isprime(p):
    for i in range(2, n):
        if n % i == 0:
            return 0
    return 1

n = int(input())
print(isprime(n))
```

```
5
1
```

```
def primeinrange(m,n):
    for i in range(m, n+1):
        if i > 1:
            for j in range(2, i):
                if i % j == 0:
                    break
            else:
                print(i)

primeinrange(2, 10)
```

```
2
3
5
7
```

```
def fact(n):
    if n == 0:
        return 1
    else:
        return n * fact(n-1)

n = int(input())
print(fact(n))
```

```
5
120
```

```
def multiplicationtable(m):
    for i in range(1, 21):
        print(f"{m} x {i} = {m * i}")

n = int(input())
multiplicationtable(n)
```

```
5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
5 x 11 = 55
```

```
5 x 12 = 60
5 x 13 = 65
5 x 14 = 70
5 x 15 = 75
5 x 16 = 80
5 x 17 = 85
5 x 18 = 90
5 x 19 = 95
5 x 20 = 100
```

```
n = 5
first, second = 0, 1

for i in range(n):
    print(first)
    first, second = second, first + second
```

```
0
1
1
2
3
```

```
def fiboseries(n):
    first, second = 0, 1
    for i in range(n):
        print(first)
        first, second = second, first + second

fiboseries(5)
```

```
0
1
1
2
3
```

```
def fib(n):
    if n == 0:
        return 0
    if n == 1:
        return 1
    else:
        return fib(n-1) + fib(n-2)
```

```
n = int(input())
print(fib(n))
```

```
6
8
```

```
def sumofn(numbers):
    total = 0
    numbers = map(int, input().split())
    for i in numbers:
        total += i
    return total

print(sumofn(a))
```

```
3 4 5
12
```

```
a = [2,3,4]
sum = 0
for i in a:
    sum = sum + i
print(sum)
```

```
9
```

```
def power(a, n):
    ans = a ** n
    return ans

a, n = map(int, input().split())
```

```
print(power(a, n))
```

```
4 4
256
```

```
normal = "123"
reverse = normal[::-1]
print(reverse)
```

```
321
```

```
num = 123
while num > 0:
    digit = num % 10
    print(digit, end = "")
    num = num // 10
```

```
321
```

```
def reverse():
    n = int(input())
    while n > 0:
        digit = n % 10
        print(digit, end = "")
        n = n // 10
    reverse()
```

```
56
65
```

```
def searchInsert(nums, target):
    for i in range(len(nums)):
        if nums[i] >= target:
            return i
    return len(nums)

nums = list(map(int, input().split()))
target = int(input())

print(searchInsert(nums, target))
```

```
44 191 209
33
0
```

```
class Solution:
    def nextGreatestLetter(letters, target):
        for ch in letters:
            if ch > target:
                return ch
        return letters[0]
print(Solution.nextGreatestLetter(letters = ["c", "f", "j"], target = "a"))
```

```
c
```

```
def factorsofn(n):
    factors = []
    for i in range(1, n):
        if n % i == 0:
            factors.append(i)
    return factors

def isprime(n):
    if n > 1:
        for j in range(2, n):
            if n % j == 0:
                return False
        return True
    return False

m = int(input())
y = factorsofn(m)
```



```
24
[2, 3]
```

```
class Solution(object):
    def search(self, nums, target):
        left = 0
        right = len(nums) - 1
        while left <= right:
            mid = (left + right)//2
            if nums[mid] == target:
                return mid
            if nums[left] <= nums[mid]:
                if nums[left] <= target < nums[mid]:
                    right = mid - 1
                else:
                    left = mid + 1
            else:
                if nums[mid] < target <= nums[right]:
                    left = mid + 1
                else:
                    right = mid - 1
        return -1
```

```
a = "c"
print(ord(a))
```

```
99
```

```
print(ord("c"))
```

```
99
```

```
for i in range(ord("A"), ord("Z") + 1):
    print(i, end = " ")
```

```
65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90
```

```
for i in range(0, 101):
    print(chr(i), end=" ")
```

```

! " # $ % & ' ( ) * + , - . / 0 1 2 3 4 5 6 7 8 9 : ; < = > ? @ A B C D E F G H I J K L M

```

```
l = list(map(int, input().split()))
for i in range(len(l)):
    for j in range(0, len(l) - 1):
        if l[j] > l[j + 1]:
            l[j], l[j + 1] = l[j + 1], l[j]
print(l)
```

```
2 3 1
[1, 2, 3]
```

```
def alphabet(n):
    vowels = []
    consonants = []
    for i in n:
        if i.lower() in "aeiou" or i.upper() in "AEIOU":
            vowels.append(i)
        else:
            consonants.append(i)
    return vowels, consonants
```

```
m = input()
vowels, consonants = alphabet(m)
print("Vowels: ", vowels)
print("Consonants: ", consonants)
```

```
Haneesh
Vowels: ['a', 'e', 'e']
Consonants: ['H', 'n', 's', 'h']
```

```
n = 6
total = 0

for i in range(1, n):
    if n % i == 0:
        total += i

if total == n:
    print("Perfect Number")
else:
    print("Not a Perfect Number")
```

Perfect Number

```
b = list(map(int, input().split()))
a = [0] + b + [0]

peaks = []

for i in range(1, len(a) - 1):
    if a[i] > a[i - 1] and a[i] > a[i + 1]:
        peaks.append(i - 1)

if len(peaks) == 0:
    print("No peaks")
else:
    largest = peaks[0]
    for i in peaks:
        if i > largest:
            largest = i
    print(largest)
```

9 6 5
0

```
class Solution(object):
    def findPeakElement(self, nums):
        left = 0
        right = len(nums) - 1
        while left < right:
            mid = (left + right) // 2
            if nums[mid] > nums[mid + 1]:
                left = mid + 1
            else:
                right = mid
        return left
```